Slide 5: table 1
Connectome_Brain_Age_Results_English_Editable.pptx

Symbol / object	Definition and dimension	How to interpret it
i; n	Scan index; number of eligible scans	A participant may contribute multiple visits; i is not a participant index.
N; A_i	N = 84 ROIs; A_i is 84 × 84	A matrix entry is connectivity between two ROIs, not a regional volume.
vec_u; x_i	Upper triangle; $x_i \in \mathbb{R}^{3486}$	Take each unordered ROI pair once; omit the diagonal.
d; X	d = N(N−1)/2 = 3,486; X is n × d	Rows are scans; columns refer to the same ROI pair across scans.
q(i); visit(i)	Participant identifier; visit identifier	q(i) defines grouping in cross-validation; visits remain together.
b(i); c_i	Technical batch; biological design vector	b(i) combines dataset and site; c_i encodes covariates to preserve.

Slide 6: table 1
Connectome_Brain_Age_Results_English_Editable.pptx

Analysis	Scans	Participants / scope
DWI harmonization	1,234	ADNI 314; HABS 487; ADRC 347; ADDECODE 86
fMRI harmonization	847	ADNI, HABS, ADRC
DWI CN OOF models	916	667 participants; HABS all-CN override
fMRI CN OOF models	642	525 participants
Paired DWI + fMRI	847	679 participants
Shared fusion CN OOF	387	330 participants

Slide 8: table 1
Connectome_Brain_Age_Results_English_Editable.pptx

Symbol / object	Definition and dimension	How to interpret it
x_{ij}; u_{ij}	Raw weight; log1p weight, dimensionless	i indexes scans; j indexes unique connections; u = log(1+x).
α_j; β_j; c_i	Edge intercept; covariate coefficients; design vector	c_i contains age and sex in the saved DWI run; diagnosis is absent.
$\gamma_{\beta j}$; $\delta_{\beta j}$	Batch location shift; positive batch scale	b = b(i) is dataset/site, not the Ridge intercept.
σ_j; ε_{ij}	Edge residual SD; standardized residual	Feature standardization precedes empirical Bayes batch estimation.
$\gamma^*_{\beta j}$; $\delta^*_{\beta j}$	Empirical Bayes batch parameter estimates	Shrink noisy batch estimates using information across connections.
H; expm1	Harmonized quantity; inverse log1p	Reconstruct symmetric weights and apply the export nonnegative convention.

Slide 10: table 1
Connectome_Brain_Age_Results_English_Editable.pptx

Symbol / object	Definition and dimension	How to interpret it
r_{ij}; z_{ij}	Correlation; Fisher-z feature	r is clipped inside (−1,1); z = arctanh(r).
c_i; β_j	Covariate design vector; edge coefficients	Age is continuous; sex and diagnosis are categorical design columns.
α_j; b(i)	Feature intercept; dataset/site batch index	Each eligible scan belongs to one batch; singleton batches are removed.
$\gamma_{\beta j}$; $\delta_{\beta j}$	Batch shift; batch scale	Adjustment removes modeled technical location and scale effects.
σ_j; γ^*; δ^*	Residual SD; empirical Bayes batch estimates	The detailed ComBat diagram shows standardization and reconstruction.
z^H; r^H	Adjusted z feature; adjusted correlation	$r^H = \tanh(z^H)$; reconstruct an 84 × 84 symmetric matrix with diagonal 1.

Slide 14: table 1
Connectome_Brain_Age_Results_English_Editable.pptx

Symbol / object	Definition and dimension	How to interpret it
X; μ; V_2	Feature matrix; column means; two PCA loadings	$T = (X-\mu)V_2$ is $n \times 2$; separate before/after fits need not share axes.
K; recall_k	Number of classes; $TP_k/(TP_k+FN_k)$	Balanced accuracy averages sensitivity equally across classes.
y_i; $\hat{y}_i$; $\bar{y}$	Actual age; estimated age; evaluation mean	Age and age errors are measured in years.
MAE; RMSE	Mean absolute error; root mean squared error	MAE = mean $\lvert\hat{y}-y\rvert$; RMSE = sqrt(mean $(\hat{y}-y)^2$).
R^2	1 − residual sum of squares / total sum of squares	$R^2 < 0$ means worse squared error than predicting the evaluation mean.
Target-specific n	Eligible scans for each diagnostic target	Sample sizes and class counts differ; do not interpret all rows as one set.

Slide 18: table 1
Connectome_Brain_Age_Results_English_Editable.pptx

Diagnostic	DWI before → after	fMRI before → after
Sex balanced accuracy	0.782 → 0.766	0.631 → 0.640
Diagnosis balanced accuracy	0.294 → 0.151	0.300 → 0.227
Age MAE (years)	6.22 → 6.71	7.64 → 8.00
Age R^2	0.588 → 0.516	0.379 → 0.307

Slide 21: table 1
Connectome_Brain_Age_Results_English_Editable.pptx

Symbol / object	Definition and dimension	How to interpret it
x_i; Z_i	Raw edge vector; standardized edge vector	Both have d = 3,486 entries; DWI and fMRI are fitted separately.
μ_j; σ_j	Training-column mean and standard deviation	Re-estimate inside every inner split; outer-test observations never fit them.
$\beta \in \mathbb{R}^d$; $b \in \mathbb{R}$	Edge coefficients; unpenalized age intercept	$\hat{y} = b + Z\beta$; β converts standardized feature values into age years.
λ; $\|\beta\|_2^2$	Nonnegative penalty; sum of squared coefficients	λ is sklearn Ridge alpha; larger values shrink coefficients more strongly.
T_o; V_o	Outer-training set; outer-test set for fold o	All visits from a participant belong to one outer fold.
λ*; OOF	Inner-CV selected penalty; out-of-fold prediction	Refit on T_o using λ*; predict V_o once. Full-data ComBat remains upstream.

Slide 24: table 1
Connectome_Brain_Age_Results_English_Editable.pptx

Symbol / object	Definition and dimension	How to interpret it
v,u; $N(v)$; E_r	ROI nodes; neighbors of v; selected edge set	Use r for retained fraction to distinguish it from a participant index.
$h_v^{(0)}$; $h_v^{(l)}$	84-value connectivity row; layer-l embedding	$h^{(0)} \in \mathbb{R}^{84}$; hidden vectors $\in \mathbb{R}^{64}$; three GINE layers.
e_{uv}; W_l	Scalar edge weight; learned edge projection	Project scalar edges to the current node-feature dimension before addition.
$m_v^{(l)}$; ε_l	Summed neighbor message; self-feature coefficient	ε is fixed at PyG default zero; the MLP transforms self + neighbor information.
g; Head(g)	64-value mean-pooled graph embedding; scalar output	Regressor: 64 → 32 → 1. Its output is standardized age.
μ_Y; σ_Y; $\hat{y}$	Training age mean / SD; predicted age in years	MSE trains standardized age; grouped inner validation selects epochs.

Slide 26: table 1
Connectome_Brain_Age_Results_English_Editable.pptx

Modality / model	CN scans	MAE (yr)	RMSE (yr)	R^2	Pearson r
DWI Ridge	916	5.98	7.89	0.519	0.724
DWI GNN (top 50%)	916	7.56	9.61	0.287	0.537
fMRI Ridge	642	7.33	9.49	0.406	0.637
fMRI GNN (top 10%)	642	9.31	11.97	0.056	0.305

Slide 29: table 1
Connectome_Brain_Age_Results_English_Editable.pptx

Symbol / object	Definition and dimension	How to interpret it
f–fold(i)	Predictor trained without scan i’s outer fold	Participant-level grouping also excludes other visits from that participant.
$\hat{y}_i$; y_i	Predicted age; chronological age, both in years	$\hat{y}$ is the raw prediction before subtracting the BAG age trend.
BAG_i	Raw brain age gap = $\hat{y}_i - y_i$	Positive values indicate older predicted age relative to chronological age.
a; b	Age-bias intercept (years); slope (years/year)	These are calibration coefficients, distinct from Ridge β and intercept.
$\hat{a}$; $\hat{b}$; $cBAG_i$	Fitted coefficients; $BAG_i - (\hat{a}+\hat{b}y_i)$	The CN OOF population defines the reference age trend.
Reference vs inference	Descriptive calibration vs independent validation	Fitting and assessing correction on one reference does not independently validate it.

Slide 32: table 1
Connectome_Brain_Age_Results_English_Editable.pptx

Symbol / object	Definition and dimension	How to interpret it
D; F	Superscripts for DWI and fMRI	Superscripts identify modalities; they are not mathematical powers.
key_i	Dataset / participant / normalized visit key	Key agreement does not establish acquisition-date proximity.
d_i; τ	Absolute age disagreement; tolerance $\tau = 0.01$ yr	Exclude unresolved ages from training; do not silently replace them.
CN^D; CN^f	Modality-specific cognitively normal indicators	DWI HABS all-CN override is retained; fMRI requires explicit CN.
$\wedge$; sex_ok	Logical AND; no explicit sex disagreement	Every required condition must hold; missing sex is allowed in this model.
n; P; $q(i)$	387 scans; 330 participants; scan-to-participant map	Pairing yields 847 scans; policy and consistency filters define the evaluated subset.

Slide 33: table 1
Connectome_Brain_Age_Results_English_Editable.pptx

Cohort	Paired scans	Evaluated CN scans	Evaluated participants
ADNI	245	131	74
ADRC	338	243	243
HABS	264	13	13
Total	847	387	330

Slide 34: table 1
Connectome_Brain_Age_Results_English_Editable.pptx

Approach	Implementation
Ridge early fusion	Concatenate standardized DWI and fMRI edges: 6,972 features
Weighted Ridge early fusion	[Z_DWI, w × Z_fMRI]; tune w and alpha jointly within training folds
Ridge late mean	Average the two unimodal age predictions
GNN fusion	Separate two-layer GINE encoders; concatenate pooled embeddings; nonlinear age head

Slide 36: table 1
Connectome_Brain_Age_Results_English_Editable.pptx

Symbol / object	Definition and dimension	How to interpret it
Z^D; Z^f	DWI and fMRI standardized edge vectors	Each has 3,486 columns; fit modality columns using training data only.
[;]; Z^E	Feature concatenation; early-fusion vector	$Z^E = [Z^D;Z^f]$ contains 6,972 features.
w; Z^W	fMRI block scale; weighted concatenation	$Z^W=[Z^D;wZ^f]$; DWI scale is one; w is not a contribution percentage.
λ*; w*	Penalty and weight jointly selected by inner CV	Search w ∈ {0,0.25,0.5,1,2,4}; λ is Ridge alpha.
β; b	6,972 edge coefficients; scalar intercept	Only β is penalized; never standardize again after block weighting.
$\hat{y}^D$; $\hat{y}^f$; ŷlate	Two unimodal OOF ages; arithmetic mean	Late fusion averages predictions in years, not coefficients or embeddings.

Slide 38: table 1
Connectome_Brain_Age_Results_English_Editable.pptx

Symbol / object	Definition and dimension	How to interpret it
$H_m^{(0)}$; m	Node input matrix; modality m ∈ {D,F}	84 × 84 input per modality; DWI log1p weights, fMRI signed correlations.
E_m; $e_{uv,m}$	Positive top-50% topology; normalized scalar weight	e = selected positive weight / graph positive maximum; bidirectional edges.
$H_m^{(l)}$; g_m	ROI embeddings; pooled modality vector	Two GINE layers, hidden 32; mean+max pooling makes $g_m \in \mathbb{R}^{64}$.
$g=[g^D;g^f]$	Concatenated fusion representation $\in \mathbb{R}^{128}$	A separate encoder is learned for each modality; weights are not shared.
Head; μ_y; σ_y	128→32→1 MLP; training age mean and SD	Head predicts standardized age; multiply by σ_y and add μ_y for age years.
$\mu_{m,j}$; $\sigma_{m,j}$	Training node-feature column statistics	Pool training graph nodes per modality; fresh outer refit follows inner epoch selection.

Slide 40: table 1
Connectome_Brain_Age_Results_English_Editable.pptx

Model	MAE (yr)	RMSE (yr)	R^2	r
DWI Ridge	6.66	8.49	0.620	0.788
Ridge early fusion	6.69	8.62	0.609	0.780
Weighted Ridge fusion	6.47	8.38	0.630	0.794
Ridge late mean	6.57	8.48	0.621	0.799
GNN fusion	9.07	11.44	0.311	0.558

Slide 41: table 1
Connectome_Brain_Age_Results_English_Editable.pptx

Cohort	CN scans	DWI Ridge MAE	Weighted Ridge MAE	GNN fusion MAE
ADNI	131	7.15	6.85	8.91
ADRC	243	6.52	6.39	9.04
HABS	13	4.12	4.16	11.24

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Connectome_Brain_Age_Results_English_Editable.pptx

Symbol / object	Definition and dimension	How to interpret it
f; x; x′	Saved outer-fold age predictor; observed input; reference	f outputs age years; x jointly contains both modalities' node and edge attributes.
x′ node / edge	Training-mean node values; zero edge attributes	Node mean corresponds to standardized zero; keep observed diagonal fixed.
k; t; x(t)	Input coordinate; path fraction; interpolated input	k is not a fold; $t \in [0,1]$; $x(t)=x'+t(x-x')$; topology is unchanged.
$\partial f/\partial x_k$; IG_k	Local model sensitivity; integrated contribution	IG has age-year units; it explains $f(x)-f(x')$, not a causal effect.
S; rIG	Integration intervals; completeness residual	Trapezoidal quadrature; S = 64 initially, adapt up to 512; $\lvert rIG\rvert \leq 0.05$ yr.
Δpred; rIG	Prediction consistency vs integration error	Require inference/checkpoint difference ≤ 0.001 yr; IG completeness is assessed separately.

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Connectome_Brain_Age_Results_English_Editable.pptx

Symbol / object	Definition and dimension	How to interpret it
$A_{s,ij}$; p; V_p	Scan connection IG; participant; participant visits	i,j now index ROI endpoints; s indexes scans to avoid an index collision.
A node; A edge	Node-profile attribution; edge-attribute attribution	Sum both symmetric node entries and both directed edge entries for each pair.
$\bar{A}_{p,ij}$; I_{ij}	Signed visit average; mean participant absolute score	Average signed scan values first, then take absolute values across participants.
ROI_i	½ $\Sigma_{j \neq i} I_{ij}$, in attribution years	Each connection contributes half its absolute importance to each endpoint.
rank; top-k frequency	Per-fold ranking; count of folds in top k	k = 100 for connections; k = 10 for ROIs; frequency ranges from 0 to 5.
x_{-m}, x'_m; ΔMAE_m	Keep other input; replace modality m; error change	Average within participant, then across participants; positive = worse prediction.

Slide 49: table 1
Connectome_Brain_Age_Results_English_Editable.pptx

Quality check	Observed result
IG integration steps	64 initially; up to 512
Completeness tolerance	0.05 years
Scans failing completeness	6 / 387 (1.6%)
Maximum absolute IG residual	0.119 years
Maximum difference from original OOF prediction	13.18 years

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Modality	Mean signed IG (yr)	Mean absolute connection IG	Occlusion error increase (yr)
DWI	-6.96	45.17	2.55
fMRI	10.33	128.54	3.36

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Connection (fMRI)	Mean |IG| (yr)	Folds in top 100
lh_precuneus ↔ rh_precuneus	0.415	5/5
lh_entorhinal ↔ rh_entorhinal	0.233	5/5
lh_isthmuscingulate ↔ lh_precuneus	0.195	5/5
lh_inferiorparietal ↔ lh_precuneus	0.183	3/5
lh_insula ↔ rh_insula	0.170	5/5

Slide 54: table 1
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Section	Primary saved evidence
Harmonization	final/Fusion/report_tables/Table3_harmonization.csv
DWI single modality	final/DWI/{Ridge_BAG,GNN_BAG}/*summary*.json / *metrics.json
fMRI single modality	final/fMRI/{Ridge_BAG,GNN_BAG}
Shared fusion benchmark	final/Fusion/report_tables/Table1,4,5,6*.csv
Interpretability	explore/GNN_attribution/*.csv; analysis_summary.json

Slide 5: table 1
Connectome_Brain_Age_Results_English_Editable_with_Pipelines.pptx

Symbol / object	Definition and dimension	How to interpret it
i; n	Scan index; number of eligible scans	A participant may contribute multiple visits; i is not a participant index.
N; A_i	N = 84 ROIs; A_i is 84 × 84	A matrix entry is connectivity between two ROIs, not a regional volume.
vec_u; x_i	Upper triangle; $x_i \in \mathbb{R}^{3486}$	Take each unordered ROI pair once; omit the diagonal.
d; X	d = N(N−1)/2 = 3,486; X is n × d	Rows are scans; columns refer to the same ROI pair across scans.
q(i); visit(i)	Participant identifier; visit identifier	q(i) defines grouping in cross-validation; visits remain together.
b(i); c_i	Technical batch; biological design vector	b(i) combines dataset and site; c_i encodes covariates to preserve.

Slide 6: table 1
Connectome_Brain_Age_Results_English_Editable_with_Pipelines.pptx

Analysis	Scans	Participants / scope
DWI harmonization	1,234	ADNI 314; HABS 487; ADRC 347; ADDECODE 86
fMRI harmonization	847	ADNI, HABS, ADRC
DWI CN OOF models	916	667 participants; HABS all-CN override
fMRI CN OOF models	642	525 participants
Paired DWI + fMRI	847	679 participants
Shared fusion CN OOF	387	330 participants

Slide 8: table 1
Connectome_Brain_Age_Results_English_Editable_with_Pipelines.pptx

Symbol / object	Definition and dimension	How to interpret it
x_{ij}; u_{ij}	Raw weight; log1p weight, dimensionless	i indexes scans; j indexes unique connections; u = log(1+x).
α_j; β_j; c_i	Edge intercept; covariate coefficients; design vector	c_i contains age and sex in the saved DWI run; diagnosis is absent.
$\gamma_{\beta j}$; $\delta_{\beta j}$	Batch location shift; positive batch scale	b = b(i) is dataset/site, not the Ridge intercept.
σ_j; ε_{ij}	Edge residual SD; standardized residual	Feature standardization precedes empirical Bayes batch estimation.
$\gamma^*_{\beta j}$; $\delta^*_{\beta j}$	Empirical Bayes batch parameter estimates	Shrink noisy batch estimates using information across connections.
H; expm1	Harmonized quantity; inverse log1p	Reconstruct symmetric weights and apply the export nonnegative convention.

Slide 10: table 1
Connectome_Brain_Age_Results_English_Editable_with_Pipelines.pptx

Symbol / object	Definition and dimension	How to interpret it
r_{ij}; z_{ij}	Correlation; Fisher-z feature	r is clipped inside (−1,1); z = arctanh(r).
c_i; β_j	Covariate design vector; edge coefficients	Age is continuous; sex and diagnosis are categorical design columns.
α_j; b(i)	Feature intercept; dataset/site batch index	Each eligible scan belongs to one batch; singleton batches are removed.
$\gamma_{\beta j}$; $\delta_{\beta j}$	Batch shift; batch scale	Adjustment removes modeled technical location and scale effects.
σ_j; γ^*; δ^*	Residual SD; empirical Bayes batch estimates	The detailed ComBat diagram shows standardization and reconstruction.
z^H; r^H	Adjusted z feature; adjusted correlation	$r^H = \tanh(z^H)$; reconstruct an 84 × 84 symmetric matrix with diagonal 1.

Slide 14: table 1
Connectome_Brain_Age_Results_English_Editable_with_Pipelines.pptx

Symbol / object	Definition and dimension	How to interpret it
X; μ; V_2	Feature matrix; column means; two PCA loadings	$T = (X-\mu)V_2$ is $n \times 2$; separate before/after fits need not share axes.
K; recall_k	Number of classes; $TP_k/(TP_k+FN_k)$	Balanced accuracy averages sensitivity equally across classes.
y_i; $\hat{y}_i$; $\bar{y}$	Actual age; estimated age; evaluation mean	Age and age errors are measured in years.
MAE; RMSE	Mean absolute error; root mean squared error	MAE = mean $\lvert\hat{y}-y\rvert$; RMSE = sqrt(mean $(\hat{y}-y)^2$).
R^2	1 − residual sum of squares / total sum of squares	$R^2 < 0$ means worse squared error than predicting the evaluation mean.
Target-specific n	Eligible scans for each diagnostic target	Sample sizes and class counts differ; do not interpret all rows as one set.

Slide 18: table 1
Connectome_Brain_Age_Results_English_Editable_with_Pipelines.pptx

Diagnostic	DWI before → after	fMRI before → after
Sex balanced accuracy	0.782 → 0.766	0.631 → 0.640
Diagnosis balanced accuracy	0.294 → 0.151	0.300 → 0.227
Age MAE (years)	6.22 → 6.71	7.64 → 8.00
Age R^2	0.588 → 0.516	0.379 → 0.307

Slide 21: table 1
Connectome_Brain_Age_Results_English_Editable_with_Pipelines.pptx

Symbol / object	Definition and dimension	How to interpret it
x_i; Z_i	Raw edge vector; standardized edge vector	Both have d = 3,486 entries; DWI and fMRI are fitted separately.
μ_j; σ_j	Training-column mean and standard deviation	Re-estimate inside every inner split; outer-test observations never fit them.
$\beta \in \mathbb{R}^d$; $b \in \mathbb{R}$	Edge coefficients; unpenalized age intercept	$\hat{y} = b + Z\beta$; β converts standardized feature values into age years.
λ; $\|\beta\|_2^2$	Nonnegative penalty; sum of squared coefficients	λ is sklearn Ridge alpha; larger values shrink coefficients more strongly.
T_o; V_o	Outer-training set; outer-test set for fold o	All visits from a participant belong to one outer fold.
λ*; OOF	Inner-CV selected penalty; out-of-fold prediction	Refit on T_o using λ*; predict V_o once. Full-data ComBat remains upstream.

Slide 24: table 1
Connectome_Brain_Age_Results_English_Editable_with_Pipelines.pptx

Symbol / object	Definition and dimension	How to interpret it
v,u; $N(v)$; E_r	ROI nodes; neighbors of v; selected edge set	Use r for retained fraction to distinguish it from a participant index.
$h_v^{(0)}$; $h_v^{(l)}$	84-value connectivity row; layer-l embedding	$h^{(0)} \in \mathbb{R}^{84}$; hidden vectors $\in \mathbb{R}^{64}$; three GINE layers.
e_{uv}; W_l	Scalar edge weight; learned edge projection	Project scalar edges to the current node-feature dimension before addition.
$m_v^{(l)}$; ε_l	Summed neighbor message; self-feature coefficient	ε is fixed at PyG default zero; the MLP transforms self + neighbor information.
g; Head(g)	64-value mean-pooled graph embedding; scalar output	Regressor: 64 → 32 → 1. Its output is standardized age.
μ_Y; σ_Y; $\hat{y}$	Training age mean / SD; predicted age in years	MSE trains standardized age; grouped inner validation selects epochs.

Slide 26: table 1
Connectome_Brain_Age_Results_English_Editable_with_Pipelines.pptx

Modality / model	CN scans	MAE (yr)	RMSE (yr)	R^2	Pearson r
DWI Ridge	916	5.98	7.89	0.519	0.724
DWI GNN (top 50%)	916	7.56	9.61	0.287	0.537
fMRI Ridge	642	7.33	9.49	0.406	0.637
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Slide 29: table 1
Connectome_Brain_Age_Results_English_Editable_with_Pipelines.pptx

Symbol / object	Definition and dimension	How to interpret it
f–fold(i)	Predictor trained without scan i’s outer fold	Participant-level grouping also excludes other visits from that participant.
$\hat{y}_i$; y_i	Predicted age; chronological age, both in years	$\hat{y}$ is the raw prediction before subtracting the BAG age trend.
BAG_i	Raw brain age gap = $\hat{y}_i - y_i$	Positive values indicate older predicted age relative to chronological age.
a; b	Age-bias intercept (years); slope (years/year)	These are calibration coefficients, distinct from Ridge β and intercept.
$\hat{a}$; $\hat{b}$; $cBAG_i$	Fitted coefficients; $BAG_i - (\hat{a}+\hat{b}y_i)$	The CN OOF population defines the reference age trend.
Reference vs inference	Descriptive calibration vs independent validation	Fitting and assessing correction on one reference does not independently validate it.

Slide 32: table 1
Connectome_Brain_Age_Results_English_Editable_with_Pipelines.pptx

Symbol / object	Definition and dimension	How to interpret it
D; F	Superscripts for DWI and fMRI	Superscripts identify modalities; they are not mathematical powers.
key_i	Dataset / participant / normalized visit key	Key agreement does not establish acquisition-date proximity.
d_i; τ	Absolute age disagreement; tolerance $\tau = 0.01$ yr	Exclude unresolved ages from training; do not silently replace them.
CN^D; CN^f	Modality-specific cognitively normal indicators	DWI HABS all-CN override is retained; fMRI requires explicit CN.
$\wedge$; sex_ok	Logical AND; no explicit sex disagreement	Every required condition must hold; missing sex is allowed in this model.
n; P; $q(i)$	387 scans; 330 participants; scan-to-participant map	Pairing yields 847 scans; policy and consistency filters define the evaluated subset.

Slide 33: table 1
Connectome_Brain_Age_Results_English_Editable_with_Pipelines.pptx

Cohort	Paired scans	Evaluated CN scans	Evaluated participants
ADNI	245	131	74
ADRC	338	243	243
HABS	264	13	13
Total	847	387	330

Slide 34: table 1
Connectome_Brain_Age_Results_English_Editable_with_Pipelines.pptx

Approach	Implementation
Ridge early fusion	Concatenate standardized DWI and fMRI edges: 6,972 features
Weighted Ridge early fusion	[Z_DWI, w × Z_fMRI]; tune w and alpha jointly within training folds
Ridge late mean	Average the two unimodal age predictions
GNN fusion	Separate two-layer GINE encoders; concatenate pooled embeddings; nonlinear age head

Slide 36: table 1
Connectome_Brain_Age_Results_English_Editable_with_Pipelines.pptx

Symbol / object	Definition and dimension	How to interpret it
Z^D; Z^f	DWI and fMRI standardized edge vectors	Each has 3,486 columns; fit modality columns using training data only.
[;]; Z^E	Feature concatenation; early-fusion vector	$Z^E = [Z^D;Z^f]$ contains 6,972 features.
w; Z^W	fMRI block scale; weighted concatenation	$Z^W=[Z^D;wZ^f]$; DWI scale is one; w is not a contribution percentage.
λ*; w*	Penalty and weight jointly selected by inner CV	Search w ∈ {0,0.25,0.5,1,2,4}; λ is Ridge alpha.
β; b	6,972 edge coefficients; scalar intercept	Only β is penalized; never standardize again after block weighting.
$\hat{y}^D$; $\hat{y}^f$; ŷlate	Two unimodal OOF ages; arithmetic mean	Late fusion averages predictions in years, not coefficients or embeddings.

Slide 38: table 1
Connectome_Brain_Age_Results_English_Editable_with_Pipelines.pptx

Symbol / object	Definition and dimension	How to interpret it
$H_m^{(0)}$; m	Node input matrix; modality m ∈ {D,F}	84 × 84 input per modality; DWI log1p weights, fMRI signed correlations.
E_m; $e_{uv,m}$	Positive top-50% topology; normalized scalar weight	e = selected positive weight / graph positive maximum; bidirectional edges.
$H_m^{(l)}$; g_m	ROI embeddings; pooled modality vector	Two GINE layers, hidden 32; mean+max pooling makes $g_m \in \mathbb{R}^{64}$.
$g=[g^D;g^f]$	Concatenated fusion representation $\in \mathbb{R}^{128}$	A separate encoder is learned for each modality; weights are not shared.
Head; μ_y; σ_y	128→32→1 MLP; training age mean and SD	Head predicts standardized age; multiply by σ_y and add μ_y for age years.
$\mu_{m,j}$; $\sigma_{m,j}$	Training node-feature column statistics	Pool training graph nodes per modality; fresh outer refit follows inner epoch selection.

Slide 40: table 1
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Model	MAE (yr)	RMSE (yr)	R^2	r
DWI Ridge	6.66	8.49	0.620	0.788
Ridge early fusion	6.69	8.62	0.609	0.780
Weighted Ridge fusion	6.47	8.38	0.630	0.794
Ridge late mean	6.57	8.48	0.621	0.799
GNN fusion	9.07	11.44	0.311	0.558

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Cohort	CN scans	DWI Ridge MAE	Weighted Ridge MAE	GNN fusion MAE
ADNI	131	7.15	6.85	8.91
ADRC	243	6.52	6.39	9.04
HABS	13	4.12	4.16	11.24

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Symbol / object	Definition and dimension	How to interpret it
f; x; x′	Saved outer-fold age predictor; observed input; reference	f outputs age years; x jointly contains both modalities' node and edge attributes.
x′ node / edge	Training-mean node values; zero edge attributes	Node mean corresponds to standardized zero; keep observed diagonal fixed.
k; t; x(t)	Input coordinate; path fraction; interpolated input	k is not a fold; $t \in [0,1]$; $x(t)=x'+t(x-x')$; topology is unchanged.
$\partial f/\partial x_k$; IG_k	Local model sensitivity; integrated contribution	IG has age-year units; it explains $f(x)-f(x')$, not a causal effect.
S; rIG	Integration intervals; completeness residual	Trapezoidal quadrature; S = 64 initially, adapt up to 512; $\lvert rIG\rvert \leq$ 0.05 yr.
Δpred; rIG	Prediction consistency vs integration error	Require inference/checkpoint difference ≤ 0.001 yr; IG completeness is assessed separately.

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Symbol / object	Definition and dimension	How to interpret it
$A_{s,ij}$; p; V_p	Scan connection IG; participant; participant visits	i,j now index ROI endpoints; s indexes scans to avoid an index collision.
A node; A edge	Node-profile attribution; edge-attribute attribution	Sum both symmetric node entries and both directed edge entries for each pair.
$\bar{A}_{p,ij}$; I_{ij}	Signed visit average; mean participant absolute score	Average signed scan values first, then take absolute values across participants.
ROI_i	½ $\Sigma_{j \neq i} I_{ij}$, in attribution years	Each connection contributes half its absolute importance to each endpoint.
rank; top-k frequency	Per-fold ranking; count of folds in top k	k = 100 for connections; k = 10 for ROIs; frequency ranges from 0 to 5.
x_{-m}, x'_m; ΔMAE_m	Keep other input; replace modality m; error change	Average within participant, then across participants; positive = worse prediction.

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Quality check	Observed result
IG integration steps	64 initially; up to 512
Completeness tolerance	0.05 years
Scans failing completeness	6 / 387 (1.6%)
Maximum absolute IG residual	0.119 years
Maximum difference from original OOF prediction	13.18 years

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Modality	Mean signed IG (yr)	Mean absolute connection IG	Occlusion error increase (yr)
DWI	-6.96	45.17	2.55
fMRI	10.33	128.54	3.36

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Connection (fMRI)	Mean |IG| (yr)	Folds in top 100
lh_precuneus ↔ rh_precuneus	0.415	5/5
lh_entorhinal ↔ rh_entorhinal	0.233	5/5
lh_isthmuscingulate ↔ lh_precuneus	0.195	5/5
lh_inferiorparietal ↔ lh_precuneus	0.183	3/5
lh_insula ↔ rh_insula	0.170	5/5

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Section	Primary saved evidence
Harmonization	final/Fusion/report_tables/Table3_harmonization.csv
DWI single modality	final/DWI/{Ridge_BAG,GNN_BAG}/*summary*.json / *metrics.json
fMRI single modality	final/fMRI/{Ridge_BAG,GNN_BAG}
Shared fusion benchmark	final/Fusion/report_tables/Table1,4,5,6*.csv
Interpretability	explore/GNN_attribution/*.csv; analysis_summary.json